სასოფლო-სამეურნეო კოოპერატივების მენეჯერების როლი კოოპერატივების განვითარებაში (საქართველოს მაგალითზე)

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რეზიუმე

სტატიაში მოცემულია სასოფლო-სამეურნეო კოოპერატივების მენეჯერების როლი კოოპერატივების განვითარებაში საქართველოს მაგალითზე დაყრდნობით. იმისათვის, რომ შეფასებულ იქნას კვლევის საგანი, გამოყენებულია, როგორც დესკრიპტული, ასევე რეგრესული ანალიზი. მონაცემები აღებულია წინასწარ შედგენილი სპეციალური კითხვარების მიხედვით 2016 წლის იანვრის თვეში. ინტერვიუები შედგა 103 სასოფლო-სამეურნეო კოოპერატივების მენეჯერთან. დესკრიპტული სტატისტიკის ანალიზზე დაყრდნობით უმეტეს შემთხვევაში სასოფლო-სამეურნეო კოოპერატივების მენეჯერებმა იციან კოოპერატივის ძირითადი პრინციპები და აგრეთვე, თუ როგორ არის გადანაწილებული პასუხისმგებლობები კოოპერატივის მენეჯერსა და დირექტორთა საბჭოს შორის. რეგრესიის მოდელის შედეგებზე დაყრდნობით შეიძლება ითქვას, რომ არსებობს დიდი შესაძლებლობა, კოოპერატივების მენეჯერებს გაუღრმავდეს ცოდნა კოოპერატივების ძირითად პრინციპებში. ამავე დროს უნდა ითქვას, რომ თუ დიდია დირექტორთა საბჭოს მოლოდინები კოოპერატივის მენეჯერებისადმი, მენეჯერებს გააჩნიათ უფრო მაღალი მოტივაცია, რათა უკეთესად გაართვან თავი თავიანთ საქმიანობას.

საკვანძო სიტყვები: სასოფლო-სამეურნეო კოოპერატივი, კოოპერატივის მენეჯმენტი.

THE ROLE OF MANAGERS OF THE AGRICULTURAL COOPERATIVES IN THE DEVELOPMENT OF COOPERATIVES (ON THE EXAMPLE OF GEORGIA)

INTRODUCTION

Cooperatives play the significant roles in agricultural development. In order to boost the farms competitiveness, one of the main roles play the agricultural cooperatives. The crop in 2002 describes that the agricultural cooperatives in USA become a significant force of agricultural sector. However, Van Niekerk (1988) emphasizes that the main reasons of agricultural cooperatives failure in South Africa are due to lack of management experiences and lack of capital resources. In addition, Akwabi-Ameyaw (1997) reports that in Africa the agricultural cooperatives often failed because of the problems in management accountable to the members (moral hazard), inappropriate political activities or financial irregularities in management.

The present study shows the agricultural cooperatives managers knowledge and their effect on the successful operation of agricultural cooperatives in Georgia. After the short introduction the paper is concentrated on the methodological part where in order to assess the research subject descriptive and regression analysis were used. After the methodology the study focuses on the data description and the data was collected based on the special survey in January, 2016. Furthermore, the paper stresses out the result and analysis. Finally, at the end it is given the conclusion remarks.

1. RESEARCH METHODOLOGY

In the methodological part it is used the quantitative research method approach. Specifically, the collected data are analyzed through descriptive statistics and consequently by the means of regression analysis. Descriptive statistics show the key tendencies of the manager's several knowledge/capabilities concerning the characteristics of the agricultural cooperative and also managers' self-assessment in the specific fields. Regression analysis models define and explain the capabilities/perceptions of the agricultural cooperative managers. The capabilities of cooperatives managers are represented by the three dependent variables denoted as GRADE1, GRADE2 and GRADE3.

In order to model the perceptions and capabilities by the means of different manager characteristics the ordinary least squares (OLS) approach is used[1]. The basic regression model is shown as follows:

$$Y = b_0 + b_1(X_1) + b_2(X_2) + b_3(X_3) + b_4(X_4) + b_5(X_5) + b_6(X_6) + b_7(X_7) + e_i$$

More specifically, regression model can be expressed in the following way:

$$\begin{aligned} GRADE = b_0 + b_1(\text{Experience}) + b_2(\text{Sales}) + b_3(\text{Education}) + b_4(\text{Age}) + b_5(\text{ManResponse}) \\ + b_6(\text{TrainingSP}) + b_7(\text{TrainingM}) + e_i \end{aligned}$$

Where $b_0 \ldots\ldots b_7$ stand for the parameters of the regression model and e_i stands for the random error term.

This empirical models are based on the research paper by Adrian and Green in 2001, where the study is conducted for the case of Alabama (USA) agricultural cooperatives. Authors' main purpose were to identify and better understand the management's role of agricultural cooperatives and business environment in Alabama.

The description of dependent variables GRADE and the set of explanatory variables are the following: **GRADE1** – is summarized/derived from the questionnaire responses and represents the manager's knowledge concerning the cooperative principles. In the model the dependent variable is ranged from 1 to 5, where 5 means the highest value;

GRADE2 – shows the manager's knowledge about manager versus board of directors' responsibilities. In addition, in the model the dependent variable is ranged from 1 to 5, where 5 denotes the highest value;

GRADE3 – is the mixed dependent variable which represents the average of GRADE1 and GRADE2. The dependent variable GRADE3 indicates manager's knowledge regarding the cooperatives principles and manager versus board of directors' responsibilities. In the model dependent variable is ranged from 1 to 5, where 5 defines the highest value;

Experience – measures the number of years working as a manager of agricultural cooperative;

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1 Software Gretl

Sales – shows sales of agricultural cooperatives in 2015, which is defined in national currency of Georgia Lari (GEL);

Education – denotes the education level which is obtained by the manager of agricultural cooperatives (Secondary education or less = 0 and tertiary education = 1);

Age – indicates the managers' age which is defined on the interview time;

ManResponse – defines the level of importance the manager pays to the board of director's perception of his/her activities (Scale is defined from 1 to 5, from which 1 is not important and 5 is extremely important);

TrainingSP – represents the value of one the fact that the manager participated in education training program in the area of structure and the principles of agricultural cooperatives. These programs are offered to the managers of agricultural cooperatives (If the manager did not participate in the training programs, the value is equal to 0 and otherwise 1);

TrainingM – measures the value of one the fact that the manager participated in education training program in the area of agricultural cooperatives management (If the agricultural cooperative manager did not participate to the management training program the value is equal to 0 and otherwise 1).

This section shows the anticipated sign of the regressor parameters of the empirical model. It shows the expected impact of the explanatory variables on the dependent variable, which is built based on the economic intuition and theoretical model.

In the empirical model all independent variables are expected to have a positive effect on the dependent variable – GRADE. The longer has the cooperative managers experience (Experience) as a manager of agricultural cooperative the better he or she become with its roles and responsibilities within agricultural cooperative.

The cooperatives size, which is shown by the sales (Sales) of agricultural cooperative in 2015, represents the flow of money in the business environment. Based on this issue it can be said that may be some programs, especially trainings in several sphere could be supported. At the same time access to the different training programs gives the opportunity to the manager to become more knowledgeable. Therefore, sales of agricultural cooperative in 2015 would positively influence on the dependent variable - GRADE.

Managers' educational level (Education) will have a positive effect on his/her ability to address/solve several tasks which is related to business environment. The cooperative manager age (Age) shows that more time and possibilities to become knowledgeable concerning the cooperative environment in the country.

The cooperative manager's response about the concerning the importance of the board of directors' perception (ManResponse) of him/her is the central point. It is known that board of directors in the agricultural cooperatives are responsible to hire, evaluate and fire the manager of the cooperative. Therefore, it is always the manager's interest to have a good relations with board of directors and ensure the agricultural cooperative is operating at its full potential.

Managers always have the incentive to expand the knowledge and management capabilities. The educational programs concerning with the structure and principles of cooperatives (TrainingSP) and management training programs (TrainingM) could develop the knowledge of managers and based on these should have a positive effect on the dependent variable.

2. DATA DESCRIPTION

2.1. Survey Analysis and Interpretation

In order to collect the cross sectional data related to the research, randomly selected managers of agricultural cooperatives were contacted via the telephone interview in the period of 12 to 25 January, 2016 in Georgia. There were approximately 1200[1] agricultural cooperatives registered in January of 2016 from which 103 agricultural cooperatives managers were interviewed.

The questionnaire was divided into five sections, which is described below in details:

In the first section of the questionnaire Rochdale principles were

1 Agricultural Cooperatives Development Agency

used to evaluate the managers' knowledge of the traditional cooperatives principles. In 1997, Bruynis, Hahn and Taylor show that the basic cooperative principles are the crucial elements for the success of the cooperatives. Therefore, if the manager does not know the basic cooperative principles and is not implementing in operation of business it has a negative effect on firm's success and operation. In this section 12 cooperatives principles are represented to the managers, where the potential response is ranged from one to five (1 – not important, 2 – slightly important, 3 – somewhat important, 4 – very important and 5 – extremely important).

In the second section of the questionnaire there are represented what is the managers' perceptions to divide the responsibilities between the manager and the board of directors which is summarized form the literature review (Mather, Ingalsbe and Volkin 1990). In 1998, Kiser in his study stressed that it is often unclear the responsibilities division between the managers and board of directors and it may create the conflicts within the cooperatives regarding the decision making issues (Cook 1994). In the second section 14 questionnaires were represented to managers and the managers response varies from one to five ranges (1 – board of directors most responsible, 2 – board of directors more responsible than manager, 3 – board of directors and manager equally responsible, 4 – manager more responsible than board of directors and 5 – manager most responsible).

The third section of questionnaire shows the managers perception of the importance of 17 different aspects to the operation and success of agricultural cooperatives. The potential response of each questioners are ranged from one to five (1 – not important, 2 – slightly important, 3 – somewhat important, 4 – very important and 5 – extremely important).

The next section (fourth) of the questionnaire shows the self-assessment of the agricultural cooperatives managers concerning the following six themes: cooperative principles, financial analysis, business decision making, strategic planning, conflict resolution and human resources management. The potential choices of the questioners are ranged from one to five (1 – poor, 2 – fair, 3 – average, 4 – good and 5 – excellent).

The final section (fifth) of questionnaire describe the characteristics of the managers and agricultural cooperatives. It includes the following issues: managers' experience, sales of cooperative in the specific time period, managers' education level and age, training program participation concerning the management, training program participation regarding cooperative structures & principles and the managers' response regarding the importance of the board of director's perception of him or her.

2.2. ANALYSIS OF DESCRIPTIVE STATISTICS

2.2.1. Characteristics of Agricultural Cooperatives and Managers in Georgia

Based on the survey agricultural cooperatives managers' ages are ranged from 25 to 78 years. The average age of agricultural cooperatives managers' is approximately 51 years. According to survey managers' experiences is ranged from 1 to 2 years. It happens because the Agricultural Cooperatives Development Agency (ACDA) was established in July, 2013 and the managers have few years of experiences as a manager of agricultural cooperatives.

According to survey the majority of agricultural cooperatives managers obtained a tertiary educational level (approximately 79 %). Managers' participation rate in the trainings regarding structure & principles of cooperatives are 77.7 %. Manager's participation in trainings concerning financial statement analysis are 65 %. In addition, 56.3 % of the managers take part in management training programs. Trainings in management, financial analysis and structure & principles of agricultural cooperatives are provided by the Agricultural Cooperatives Development Agency (ACDA) and several international donor organizations.

2.2.2. Principles of Cooperatives

In the table 1 there are twelve key cooperatives principles listed and for each principle the managers' responds are defined in percentages. Ten out of twelve cooperatives principles were classified as very or extremely important for the success and operation of the agricultural cooperatives. According to the data collected “very” or “extremely” important principles are follows: equity is provided by patrons/owners (97.1 % managers responded by the value 4 or 5), membership is open (96.1 % managers responded by the value 4 or 5), net income is allocated to patrons as patronage refunds (96.1 % managers responded by the value 4 or 5) and have a duty to educate members (96.1 % managers responded by the value 4 or 5).

Table 1. Managers' rankings of importance of cooperative principles to the operation and success of agricultural cooperatives

		Ranking				
		1	2	3	4	5
1	One-member one-vote	1.9%	0.0%	11.7%	40.8%	45.6%
2	Membership is open	1.0%	0.0%	2.9%	53.4%	42.7%
3	Equity is provided by patrons/owners	0.0%	0.0%	2.9%	61.2%	35.9%
4	Equity ownership share is limited for each single number	29.1%	5.8%	28.2%	31.1%	5.8%
5	Net income is allocated to patrons as patronage refunds	1.0%	0.0%	2.9%	55.3%	40.8%
6	Dividend on equity capital is limited	5.8%	0.0%	14.6%	52.4%	27.2%
7	Exchange of goods and services is at market price	1.9%	0.0%	5.8%	65.0%	27.2%
8	Have a duty to educate members	1.0%	0.0%	2.9%	46.6%	49.5%
9	Use of cash trading only	34.0%	5.8%	32.0%	22.3%	5.8%
10	No unusual risk is assumed	0.0%	0.0%	16.5%	71.8%	11.7%
11	Maintain political and religious neutrality	6.8%	1.9%	9.7%	50.5%	31.1%
12	Have equality of the sexes in membership	2.9%	1.9%	2.9%	58.3%	34.0%
Number of observations - 103						
Responses are ranged from one to five: 1 - not important; 2 - slightly important; 3 - somewhat important; 4 - very important; 5 - extremely important.						

Source: Based on Survey & Author’s Calculation

2.2.3. Agricultural Cooperatives Manager vs Board of Directors Responsibilities

In the table 2, it is represented fourteen questionnaires concerning how the managers' and board of directors' responsibilities are divided with possible responses. Each of the responses are given in percentages. In majority of questionnaires managers somehow correctly answered about their responsibilities as it is determined by the Meyer in 1994. According to the table 2 the key areas where managers understand their responsibilities are: managing the day-to-day operations of the cooperative with the 69.9 %, informing members of developments within the cooperative (49.5 %), educating the general public about the cooperative and its activities (48.5 %), ensuring employees understand cooperative philosophy (47.6 %), furnishing information needed for long range planning (40.8 %) and acting in good faith with reasonable care in handling the affairs of the cooperative (35 %). However, managers failed to identify their role in the following issue - developing programs for implementation of cooperative policies (16.5 %).

Table 2. Managers' response for division of responsibilities between management and the board of directors

		Ranking				
		1	2	3	4	5
1	Setting the direction of the business for the welfare of the cooperative members	26.2%	9.7%	44.7%	3.9%	15.5%
2	Managing the day-to-day operations of the cooperative	3.9%	2.9%	23.3%	1.9%	68.0%
3	Maintaining accuracy of the minutes of the board of directors' meetings	11.7%	7.8%	27.2%	13.6%	39.8%
4	Acting in good faith with reasonable care in handling the affairs of the cooperative	11.7%	1.9%	51.5%	13.6%	21.4%
5	Ensuring employees understand cooperative philosophy	9.7%	7.8%	35.0%	13.6%	34.0%
6	Approving purchase of major capital assets	50.5%	11.7%	27.2%	4.9%	5.8%
7	Developing programs for implementation of cooperative policies	33.0%	5.8%	44.7%	7.8%	8.7%

8	Establishment and evaluation of programs	28.2%	3.9%	43.7%	12.6%	11.7%
9	Furnishing information needed for long range planning	18.4%	5.8%	35.0%	17.5%	23.3%
10	Educating the general public about the cooperative and its activities	14.6%	2.9%	34.0%	18.4%	30.1%
11	Keeping current on legislation concerning cooperatives	14.6%	3.9%	37.9%	9.7%	34.0%
12	Encouraging membership and active patronage	21.4%	6.8%	35.0%	8.7%	28.2%
13	Informing members of developments within the cooperative	16.5%	2.9%	31.1%	16.5%	33.0%
14	Hiring, training, and setting compensation for employees	24.3%	5.8%	37.9%	9.7%	22.3%
Number of observations - 103						
Responses are ranged from one to five: 1 - board is most responsible; 2 - board more responsible; 3 - board and manager equally responsible; 4 - manager more responsible; 5 - manager most responsible.						

Source: Based on Survey & Author's Calculation

Agricultural cooperative managers somehow understand the board of directors' responsibilities: approving purchase of major capital assets (62.1 %), setting the direction of the business for the welfare of the cooperative members (35.9 %) and establishment and evaluation of programs (32 %). However, manager failed in identifying the following board of directors' role – maintaining accuracy of the minutes of the board of directors' meetings (19.4 %).

2.2.4. Agricultural Cooperative Management

In the table 3 there are represented seventeen questionnaires regarding the several management issues which is important for success and operation of the agricultural cooperatives. Each of the responses are given in percentages. Majority of the managers in the selected management questionnaires responded that all issues are "very" and "extremely" important. Cooperative members', board of directors' and employees' perceptions of management are categorized by 98.1 %, 97.1 % and 85.4 % correspondingly as being "very important" and "extremely important". In addition, managers' emphasized that cooperatives members' perception of the board of directors is very and extremely important with the 96.1 %.

For the managers written policy procedures and job description are very and extremely important with the 96.1 % and 95.1 % respectively. For the operation and success of the agricultural cooperatives preventive maintenance programs for buildings & warehouses and preventive maintenance programs for equipment are very and extremely important with the 92.2 % and 94.2 % respectively.

Managers are identified that management's knowledge of products/services offered by the cooperative (99 %) and employees' knowledge of products/services offered by the cooperative (86.4 %) are very and extremely important for the operation and succession of the agricultural cooperatives. In addition, managers clearly identified that effective employee incentive program (92.2 %) and advanced training for employees (93.2 %) play the crucial role in operation and success of the agricultural cooperatives.

Table 3. Managers response for management issues for the success of the cooperatives

		Ranking				
		1	2	3	4	5
1	Cooperative members' perception of management	1.0%	0.0%	1.0%	55.3%	42.7%
2	Board members' perception of management	0.0%	0.0%	2.9%	53.4%	43.7%
3	Employees' perception of management	2.9%	1.9%	9.7%	50.5%	35.0%
4	Cooperative members' perception of the board of directors	0.0%	0.0%	3.9%	62.1%	34.0%
5	Use of budgets within each department	1.0%	0.0%	5.8%	68.0%	25.2%
6	Financial analysis within each department	0.0%	0.0%	3.9%	68.0%	28.2%
7	Written policies/procedures	0.0%	0.0%	3.9%	61.2%	35.0%
8	Written job descriptions	0.0%	0.0%	4.9%	56.3%	38.8%

9	Established performance goals and/or standards	1.0%	1.0%	4.9%	51.5%	41.7%
10	Preventive maintenance programs for buildings and warehouses	1.0%	0.0%	6.8%	59.2%	33.0%
11	Preventive maintenance programs for equipment	0.0%	0.0%	5.8%	63.1%	31.1%
12	Management's knowledge of products/services offered by the cooperative	0.0%	0.0%	1.0%	31.1%	68.0%
13	Employees' knowledge of products/services offered by the cooperative	0.0%	1.9%	11.7%	49.5%	36.9%
14	Introducing/researching new products	1.9%	0.0%	6.8%	57.3%	34.0%
15	Employees' understanding of cooperative principles	4.9%	1.0%	15.5%	56.3%	22.3%
16	Effective employee incentive program	1.0%	1.0%	5.8%	65.0%	27.2%
17	Advanced training for employees	1.9%	0.0%	4.9%	47.6%	45.6%
Number of observations - 103						
Responses are ranged from one to five: 1 - not important; 2 - slightly important; 3 - somewhat important; 4 - very important; 5 - extremely important.						

Source: Based on Survey & Author's Calculation

2.2.5. Agricultural Cooperative's Managers Self-Assessment

The table 4 represents the agricultural cooperative managers' response about the self-assessment in several management topics. Each of the responses are given in percentages. The average response of managers' self-assessment is ranged from 3 to 3.9 in the 1 to 5 scale, where 1 is poor and 5 is excellent assessment criteria. The lowest average score was noted in financial analysis – 3 and the highest average score was used in conflict resolution – 3.9. In addition, according to the data and self-assessment responses the managers are „good" and „excellent" in conflict resolution (82.5 %) and human resource management (77.7 %). However, manager's self-assessment are poor and fair in financial analysis with the 25.2 %.

Table 4. Managers' self-assessment in management knowledge/capabilities						
		Ranking				
		1	2	3	4	5
1	Cooperative Principles	1.9%	9.7%	36.9%	44.7%	6.8%
2	Financial Analysis	3.9%	21.4%	47.6%	26.2%	1.0%
3	Business Decision Making	2.9%	8.7%	30.1%	54.4%	3.9%
4	Strategic Planning	1.0%	11.7%	40.8%	43.7%	2.9%
5	Conflict Resolution	0.0%	7.8%	9.7%	65.0%	17.5%
6	Human Resource Management	0.0%	6.8%	15.5%	67.0%	10.7%
Number of observations - 103						
Responses are ranged from one to five: 1 - poor, 2 - fair, 3 - average, 4 - good, and 5 - excellent.						

Source: Based on Survey & Author's Calculation

3. RESULTS AND DISCUSSIONS

In order to analyze the managers and cooperatives characteristics and their effect on either managers' response about the cooperatives principles (GRADE1) in the first model or on manager vs board of directors' responsibilities (GRADE2) in the second model or on the mixed response (GRADE3) in the last model, the regression analysis are conducted.

Ordinary Least Square (OLS) estimation method is used. In the classical linear regression model the residuals should be normally distributed. At the same time there should not be multicollinearity and heteroscedasticity problems in the empirical model. The number of observations is 103 as 103 managers of agricultural cooperatives filled the questionnaires. When developing models the assumptions are tested first. The tables 5, 6, and 7 and 8 present results associated with model assumptions: no multicollinearity, no heteroscedasticity and normal distribution of residuals.

In order to assess whether there is the multicollinearity assumption violation it is used the Variance Inflation Factors (VIF), which is shown on the table 5. Based on the VIF values it can be concluded that there is no multicollinearity problem in the empirical model, because all the VIF values of all independent variables are below the 10.

Table 5. Variance Inflation Factors (VIF)	
Experience	1.039
Sales	1.077
Education	1.067
Age	1.061
ManResponse	2.078
TrainingSP	1.674
TrainingM	1.739
Minimum possible value = 1.0 Values > 10.0 may indicate a collinearity problem	

Source: Author's Calculation

In order to test for the residual normality Jarque-Bera test is used. According to the test result p-value is extremely low in table 9 for model 1 and there is the residual normality problem ($p=2.7144e\text{-}006<0.05=\alpha$). In order to test the heteroscedasticity it is used the Brauch-Pagan test and based on the result the p-value is very low ($p=0.000008<0.05=\alpha$). So, there is the problem of heteroscedasticity in the model 1. Therefore, model 1 in table 9 failed in both homoscedasticity and residual normally distribution assumptions.

Table 6. Tests for Model 1		
JB test	Jarque-Bera test = 25.6339	p-value = 2.7144e-006
Brauch-Pagan test	LM = 35.814224	p-value = 0.000008

Source: Author's Calculation

According to the p-value results of JB test is extremely low in model 2 table 9 and there is the residual normality problem in the empirical model (p=0.0255287<0.05=α). In order to test the heteroscedasticity it is used the Brauch-Pagan test and based on the result the p-value is high (p=0.531730>0.05=α). Therefore, there is no problem of heteroscedasticity in the model, however residuals are not normally distributed.

Table 7. Tests for Model 2		
JB test	Jarque-Bera test = 7.33591	p-value = 0.0255287
Brauch-Pagan test	LM = 6.068877	p-value = 0.531730

Source: Author's Calculation

In table 9 model 3, JB test results and p-value is high enough compared with α significance level which is equal to 0.05. Thus it can be concluded that residuals are normally distributed (p=0.23>0.05=α). The p-value in the Brauch-Pagan test result is very high (p=0.835>0.05=α). So, it suggests that there is no problem of heteroscedasticity in the model 3. Therefore, there is no heteroscedasticity problems and residual are normally distributed in model 3 table 9. So, based on the tests results and VIF values the model 3 in table 9 fits to the classical linear model assumptions and the regression results can be interpreted.

Table 8. Tests for Model 3		
JB test	Jarque-Bera test = 2.90876	p-value = 0.233546
Brauch-Pagan test	LM = 3.498253	p-value = 0.835411

Source: Author's Calculation

Table 9. Regression Model and Results			
	Ordinary Least Square (OLS)		
	Model 1	**Model 2**	**Model 3**
Variables	GRADE1	GRADE2	GRADE3
Constant	2.34250 (0.172564) ***	2.79294 (0.355768) ***	2.56772 (0.20798) ***
Experience	0.0202689 (0.0395074)	−0.0499309 (0.081451)	−0.014831 (0.0476158)

Sales	−2.79030e-07 (1.25602e-07) **	1.62314e-07 (2.5895e-07)	−5.8358e-08 (1.51381e-07)
Education	−0.0540660 (0.0462545)	−0.0352519 (0.0953613)	−0.0446589 (0.0557477)
Age	2.99270e-05 (0.00164286)	0.00194568 (0.00338702)	0.000987803 (0.00198003)
ManResponse	0.290204 (0.0412061) ***	0.141184 (0.0849531) *	0.215694 (0.0496631) ***
TrainingSP	0.284015 (0.0570255) ***	−0.18938 (0.117567)	0.0473176 (0.0687293)
TrainingM	0.233777 (0.0488059) ***	−0.0806556 (0.100621)	0.0765607 (0.0588227)
R-square	0.756847	0.044747	0.383598
Adjusted R-square	0.738931	-0.025640	0.338179
P-value (F)	1.75e-26	0.725270	5.29e-08
Number of Observations	103	103	103
*** Significant at the 1 % level. * Significant at the 10 % level.	** Significant at the 5 % level. Standard errors in parentheses.		

Source: Author's Calculation

The regression results are presented on the table 9. In the table 9 model 3, adjusted R-square is acceptable for the cross-sectional data to explain the regression model and it equals to approximately 34 %. In the model 3, intercept and one independent variable are found to be significant - manager's concern with the board's perception of him/her (ManResponse). The related p-value is 0.0001 and is significant at 1 % significance level. As expected Manager's concern with the board's perception of him/her (ManResponse) has a positive effect on the dependent variable. If the managers' concern to the board of directors' increased by 1 unit their grade on average will increase by approximately 0.22 points if the other variables are equal.

In order to improve the first and second model in table 9, because of violation of basic assumptions, the log form of the Sales are used. For that reason the cases where sales of agricultural cooperatives are equal to zero are excluded. The sample size then is 70 cases which is shown in the table 14. As a consequence, the model represents only that part of agricultural cooperatives which have positive sales in the country. In other words it means that if the cooperative has zero sales in 2015 it means that it does not operate in the market. After transforming sales variable in natural logarithmic values the regression model assumptions should be tested again.

As it is known in order to test the multicollinearity it is used the Variance Inflation Factors (VIF) test. Based on the test results it can be conclude that there is not any collinearity problems in the model, because the all values of independent variables are below the 10.

Table 10. Variance Inflation Factors (VIF)	
Experience	1.114
LogSales	1.127
Education	1.150
Age	1.111
ManResponse	1.912
TrainingSP	1.845
TrainingM	1.863
Minimum possible value = 1.0 Values > 10.0 may indicate a collinearity problem	

Source: Author's Calculation

In order to test for the residual normality it is used Jarque-Bera test in each of the three models. According to the test results p-values are high enough and its equal to approximately 0.7, 0.23 and 0.31 properly, which is shown in tables 11, 12 and 13. So, p-values are more than 0.05=α. Hence, the residuals are normally distributed in the each of the three models. Thus, after transformation the residuals became normally distributed in each three models in table 14.

In order to test the heteroscedasticity it is used the Brauch-Pagan test in each of the three models in table 14. According to the results the p-values are high enough and equal to approximately 0.11, 0.7 and 0.8

appropriately, which is shown in tables 11, 12 and 13. So, p-values are more than 0.05=α. Therefore, after transforming the model there is no heteroscedasticity problem in each of the three models in table 14.

Table 11.Tests for Model 1		
JB test	Jarque-Bera test = 0.714432	p-value = 0.699621
Brauch-Pagan test	LM = 11.649461	p-value = 0.112692

Source: Author's Calculation

Table 12. Tests for Model 2		
JB test	Jarque-Bera test = 2.91515	p-value = 0.232801
Brauch-Pagan test	LM = 4.543992	p-value = 0.715414

Source: Author's Calculation

Table 13. Tests for Model 3		
JB test	Jarque-Bera test = 2.371	p-value = 0.305593
Brauch-Pagan test	LM = 3.625560	p-value = 0.821753

Source: Author's Calculation

After transformation of the model it can be concluded that there is no collinearity and heteroscedasticity problems in each of the models and the residuals are normally distributed in table 14. Therefore, after transformation each regression models and variables can be interpreted properly.

Table 14. Regression Models and Results			
	Ordinary Least Square (OLS)		
	Model 1	**Model 2**	**Model 3**
Variables	GRADE1	GRADE2	GRADE3
Constant	2.44691 (0.231465) ***	2.50733 (0.603193) ***	2.47712 (0.344153) ***
Experience	0.0200775 (0.0443537)	0.104613 (0.115585)	0.0623451 (0.0659471)
LogSales	−0.0356707 (0.0144247) **	0.0309175 (0.0375904)	−0.00237656 (0.0214473)

Education	−0.0506954 (0.0503236)	−0.063127 (0.131142)	−0.0569112 (0.0748234)
Age	−0.000128313 (0.00193590)	0.00328968 (0.00504492)	0.00158068 (0.00287839)
ManResponse	0.351942 (0.0428129) ***	0.0729497 (0.111569)	0.212446 (0.0636562) ***
TrainingSP	0.249170 (0.0637601) ***	−0.255667 (0.166157)	−0.00324874 (0.0948015)
TrainingM	0.224023 (0.0529166) ***	−0.131087 (0.137899)	0.046468 (0.0786788)
R-square	0.815122	0.053356	0.326238
Adjusted R-square	0.794249	-0.053523	0.250168
P-value (F)	2.12e-20	0.831560	0.000638
Number of Observations	70	70	70
*** Significant at the 1 % level. ** Significant at the 5 % level. * Significant at the 10 % level. Standard errors in parentheses.			

Source: Author’s Calculation

In the table 14 model 1, adjusted R-square is high to explain the GRADE1 variable and it’s equal to approximately 79 %. In the mode 1, intercept and four variables are found to be significant: sales of the co-operatives in 2015 (LogSales), manager’s concern with the board’s perception of him/her (ManResponse), participation in the training program concerning cooperatives structures and principles (TrainingSP) and participating in the trainings regarding the management (TrainingM). One variable, namely - LogSales out of these four variables surprisingly has a negative effect on the dependent variable. The related p-value of sales is 0.0162 and is significant at 5 % significance level. If the sales of the agricultural cooperatives increased by 1 percent the dependent variable GRADE1 on average will decrease by 0.04 points in ceteris paribus. Manager’s concern with the board’s perception of him/her (ManResponse), participation in the training program concerning cooperatives structures and principles (TrainingSP) and participating in the trainings in management (TrainingM) variables as expected have a positive effect on the dependent variable with the related p-values 0.0001, 0.0002 and 0.0001 properly and all are significant at 1 % significance level. If the managers' concern to the board of directors' increased by 1 unit their grade on average will increase by 0.35 points in ceteris paribus. If the managers' participated in the training concerning cooperatives structure and principles the grade will increase on average by 0.25 points when the other factors are equal. If the managers' participated in the training regarding management the grade will increase on average by 0.22 points when the other factors are equal.

In the table 14 model 2, adjusted R-square is extremely low in order to explain the regression model. In the model 2 only the intercept is found to be significant with the 1 % level.

In the table 14 model 3, adjusted R-square is acceptable for the cross-sectional data to explain the regression model and it equals to approximately 25 %. In the model 3, intercept and only one independent variable is found to be significant - manager's concern with the board's perception of him/her (ManResponse). The related p-value is 0.0014 and is significant at 1 % level. Manager's concern with the board's perception of him/her (ManResponse) has a positive effect on the dependent variable. If the managers' concern to the board of directors' increased by 1 unit their grade on average will increase by approximately 0.21 points in ceteris paribus.

4. CONCLUSION

According to the survey analysis in the majority cases managers somehow understand the cooperative principles. Based on the data analyzed the managers got highest scores in the following four cooperative principles: a) equity is provided by patrons/owners b) membership is open c) net income is allocated to patrons as patronage refunds d) have a duty to educate members.

Based on the survey analysis in the majority cases managers somehow understand the manager's vs board of directors responsibilities. The key area where the managers more or mostly understand their responsibility with highest value is managing the day-to-day operations of the cooperative. However, managers failed to identify their role in the following issue - developing programs for implementation of cooperative policies. Furthermore, the main area where the managers more or mostly understand the board of directors' responsibility with highest value is approving purchase of major capital assets of cooperative. However, manager failed in identifying the following board of directors' role – maintaining accuracy of the minutes of the board of directors' meetings.

The majority of the managers in the selected management questionnaires respond that all management topics are "very" and "extremely" important. The key very or extremely important management topics, which are identified by the managers are - management's knowledge of products/services offered by the cooperative and cooperative members' perceptions of management.

In the managers self-assessment survey the highest average score was performed in "conflict resolution" and the lowest average score was performed in "financial analysis". As all self-assessment variables are measured at the same scores it can be concluded that agricultural cooperative managers have the highest expertise in "conflict resolution" and the lowest expertise in cooperative's "financial analysis".

The managers of agricultural cooperatives perfectly understand the roles of selected management issues such as the written policies & procedures, effective employee incentive programs, advanced training programs, management knowledge of products and services offered by the cooperative. These issues are crucial for the successful operation of agricultural cooperatives. In addition, managers realize and recognize that cooperative members, board of directors and employee perception of management are substantially important for the success of agricultural cooperatives in Georgia.

Initially, 103 agricultural cooperatives are taken from the total number of cooperatives. From the regression model 3, as expected, it is found that variable the managers' concern with the board's perception of him/her (ManResponse) has a significantly positive effect on dependent variable - GRADE3. It could be said that managers care for the board of director's attitude to him/her. In addition, if there are the high expectations coming from the board of director's managers can handle their tasks better which leads to successful operation of the agricultural cooperatives in Georgia.

The final model represents only the part of agricultural cooperatives (70 agricultural cooperatives) which have positive sales in Georgia. In other words it means that if the cooperative has zero sales in 2015 it means it does not operate in the market. Initially, in the regression model it is expected that all of the independent variables should have the positive impact on the dependent variables. Surprisingly, in the model 1 the sales of the agricultural cooperatives (LogSales) had a significant negative effect for the cooperatives principles GRADE1. This surprising result should be further investigated in future research.

In the final model 1, manager's concern with the board's perception of him/her (ManResponse), participation in the training program concerning cooperatives structures and principles (TrainingSP) and participating in the trainings regarding in management (TrainingM) as expected have a positive effect on the manager's knowledge concerning cooperatives principles.

In the final model 3, manager's concern with the board's perception of him/her (ManResponse) has the positive effect on the mixed grade of manager's knowledge about the cooperatives principles and the managers versus to board of director's responsibilities.

Results of the regression models implies that there are an opportunities to broaden the managers' knowledge/capabilities concerning the cooperative principles and management. In other words it means that training programs in structure & principle and in management substantially positively affect managers' performance in Georgia. At the same time better skilled managers substantially positively affect the success and operation of the agricultural cooperatives in the country.

From the regression models it is found that variable the managers' concern with the board's perception of him/her (ManResponse) has a significantly positive effect on dependent variable - GRADE. In other words it could be said that managers can cope their tasks better if there are high expectations coming from the board of directors of the agricultural cooperatives. Therefore, manager's better performance caused by the board of director's expectations lead as a success and proper operation of the agricultural cooperatives in the country. Based on the research if there is higher expectation of boar d of directors to managers, the managers have higher intention to perform better.

This research will be the step forward in broadening the knowledge in agricultural cooperatives area which may help to develop the policy making and consequently may support decision makers to address the problems faced in agricultural cooperation in Georgia.

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